INDEPENDENT FLAGSHIP-CLASS PROPOSAL · ICLR 2027

From Fragments to an Executable Passport

Coherent probabilistic lineage, impact, loss, recall, and circularity from incomplete and contradictory supply-chain evidence.

LANTERN-SC

Lineage-Aware Network for Transformation Evidence, Reconciliation, and Non-identifiability in Supply Chains

Evidence cutoff: 30 August 2026 · Standalone third proposal · Sustainable and resilient supply chains

STANDALONE RESEARCH DECISION

Executive research case

A product passport should not merely store claims. It should answer lineage, footprint, loss, recall, and recovery questions from one physically coherent model—and state exactly what the available evidence cannot determine.

Central thesis

LANTERN-SC learns an evidence-constrained posterior over split, merge, transformation, loss, and return events. Every sustainability answer is a query on that same posterior; every answer carries provenance, calibrated uncertainty, and a reconciliation certificate.

Real failure

Passports and PCFs can be complete in syntax while the physical history is missing, conflicting, or false.

Scientific object

A typed stochastic transformation hypergraph with evidence and allocation semantics.

Novelty nucleus

Sharp partial identification plus cross-query coherence under hidden split/merge histories.

One-stop value

Traceability, carbon, waste, recall, circularity, anomaly localization, and next-best evidence.

8.5/10

Novelty ceiling

High only if transformation identity and partial-identification theory exceed ordinary flow completion.

9/10

Umbrella unity

Multiple workflows are queries on one latent physical history—not a bundle of unrelated agents.

10/10

Domain fit

Directly aligned with DPP, Scope 3 data quality, circularity, and multi-tier traceability.

HONEST STATUS

This is a **conditional GO**, not a novelty guarantee. If the central theorem reduces to a standard incidence-matrix rank test, or realistic evidence yields vacuous intervals, the flagship claim fails.

Contributions that form one paper

C1 · Object
transformation hypergraph

C2 · Theory
identification and sharp bounds

C3 · Engine
coherent evidential queries

C4 · Artifact
SC-PASSPORT-Bench

01 · THE MOTIVATING FAILURE

One battery, six incompatible truths

A battery pack arrives with a valid QR code and signed declarations. The identifiers are real. The numbers still need not describe what physically happened.

Observed fragments

- mine-level certificate for a precursor batch;
- supplier invoice with aggregate monthly mass;
- cell-plant electricity meter and production count;
- pack-level EPCIS aggregation event;
- transport PCF calculated under one method version;
- recycler claim for recovered nickel and black mass.

Hidden physical history

- two precursor lots were blended;
- one transformation event is absent;
- yield loss was recorded as generic waste;
- the same recycled-content claim appears twice;
- a supplier changed its emission factor mid-period;
- returned modules were split across reuse and recycling.

Independent models disagree by construction

Question	Typical isolated system	Failure
Which upstream lots entered this pack?	traceability database	missing transformation is silently treated as no link
What is its product carbon footprint?	PCF calculator	allocates aggregate energy without lineage uncertainty
Which packs must be recalled?	rule or graph traversal	returns a brittle yes/no set from uncertain ancestry
Where did mass disappear?	waste dashboard	does not reconcile with output lots and inventory
Can this module be reused?	circularity classifier	ignores uncertain chemistry, condition, and prior transformations
Which document should be audited?	manual judgment	does not target the decision-critical uncertainty

LANTERN ANSWER

Return a joint posterior and sharp query intervals: pack A contains 0.42–0.57 kg of precursor lot X; its cradle-to-gate footprint is 81–96 kg CO_2e under method version v; recall exposure is 0.73; and verifying one hidden blending record would shrink both lineage and footprint uncertainty most.

The ecosystem has standards—not an inference layer

Current standards solve essential pieces of syntax, accounting, exchange, and access. None converts incomplete multi-enterprise evidence into a calibrated, physically reconciled history.

Layer	What it supplies	Remaining research gap
GHG Protocol Scope 3[1]	corporate value-chain inventory and data hierarchy	supplier gaps, data specificity, uncertainty, and non-causal attribution
PACT v3[2]	method and exchange format for product carbon footprints	truth of primary evidence, missing transformations, coherent non-carbon queries
ESPR / DPP[3]	product identifier, access, registry, product-group data	a passport is a container; it does not infer an uncertain physical history
GS1 EPCIS 2.0[4]	object, aggregation, transaction, association, and transformation events	inference when events are absent, duplicated, inconsistent, or strategically hidden
UNTP / credentials[5]	traceability events, passports, and issuer provenance	a signed claim can still be wrong about the physical world
ISO 59020 / PEF[6]	circularity measurement and lifecycle impact method	latent material closure, uncertain recovery, and impact reconciliation
EU Data Act[7]	access, interoperability, and trade-secret safeguards	what minimum aggregate disclosure still permits useful identification

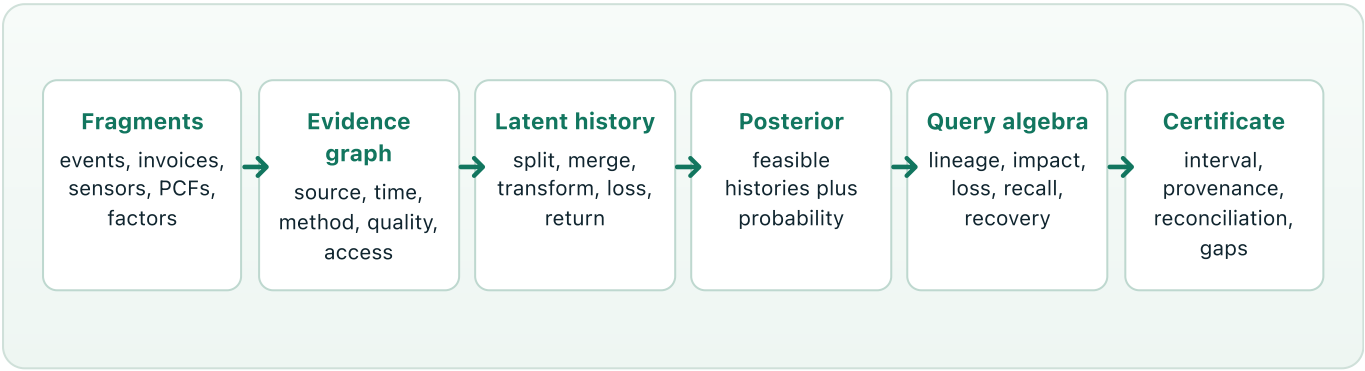
THE PRECISE WEDGE

Standards say how a claim should be represented. LANTERN asks whether the set of claims can jointly correspond to any physical history, which histories remain possible, and which sustainability answers are invariant across them.

Regulatory status is time-sensitive. The artifact must freeze exact versions; proposed 2026 Scope 3 revisions must not be presented as final requirements.[8]

One scientific object, seven downstream queries

The project is an umbrella because one latent history supports many questions—not because seven modules are placed behind one interface.



Traceability

Which input lots may or must be ancestors of this product?

Carbon & impacts

What range is consistent with lineage, factors, allocation, and boundary?

Recall

Which downstream items are exposed, with what probability and worst-case set?

Loss & waste

Where can unobserved scrap, spoilage, or leakage be physically located?

Circular recovery

Which reuse, repair, remanufacture, or recycling claims are feasible?

Anomaly & audit

Which evidence conflict explains infeasibility, and what should be checked next?

BOUNDARY

Labor conditions, human rights, and all ESG concerns are **not** physically conserved. They may enter as separately evidenced attributes, but LANTERN must not claim that mass-balance theory solves them.

What the model receives—and may never assume

Observed evidence Y

- partial EPCIS-style events and identifiers;
- invoices, manifests, bills of material, and declarations;
- sensor intervals and condition readings;
- facility totals, production counts, inventory snapshots;
- emission factors, PCFs, method/boundary versions;
- verification status, source identity, timestamps, access policy.

Latent variables H

- unobserved events, lots, and parentage;
- material quantities, qualities, and transformation yields;
- loss, scrap, spoilage, and recovered flows;
- impact attribution through transformations;
- evidence reliability or contamination state;
- which feasible history generated each query answer.

Assumptions must be machine-readable

Assumption	Declared input	Never silently inferred	Stress test
Boundary	cradle-to-gate, gate-to-gate, lifecycle stage	omitted tiers or capital goods	expand/contract boundary
Allocation	mass, energy, economic, substitution rule	the “best” rule from outcomes	allocation sensitivity
Transformation	allowed input/output types and yield ranges	chemically impossible substitutions	misspecified grammar
Missingness	random, tiered, strategic, or unknown class	missing-at-random convenience	MNAR and adversarial
Evidence trust	issuer, method, verification, source class	signature equals truth	collusion and corruption

ABSTENTION RULE

If no feasible history exists, return a minimal conflict set. If many histories exist and the query interval is too wide for the intended decision, abstain and rank additional evidence. Never hide non-identifiability behind a point estimate.

05 · MATHEMATICAL OBJECT

A typed stochastic transformation hypergraph

Let **G = (V,E,τ,η)** be a temporal directed hypergraph. Vertices are lots, products, facilities, actors, and evidence records. A hyperedge **e: I_e → O_e** maps multiple input lots to multiple outputs through a typed transformation.

$$p_\theta(X, L, C, W, R, Z \mid Y, G, D, \nu, B) \propto \mathbb{1}[\text{typed physical constraints}] \cdot p_\theta(\text{history} \mid G) \cdot \prod_{g \in D} \psi_g(Y_g \mid \text{history}, \text{sources}, \text{methods})$$

Symbol	Meaning	Required structure
X	material quantities, lot states, qualities	nonnegative, typed units, bounded conversion
L	lineage fractions and ancestry	split/merge identity, no impossible parentage
C	impact-vector contributions	method and lifecycle boundary attached
W	loss, scrap, spoilage, recovered material	explicit sink/source and disposition
R	reliability or contamination indicators	source-aware, not universal trust score
Z	claim-to-evidence provenance	which records support each inferred fact
D	evidence dependency/copying clusters	prevents duplicated sources from creating false precision
ν, B	allocation/method version and boundary	frozen per answer; never implicit

Why typed transformations matter

Merge

Several lots blend into one output; ancestry is fractional rather than a single edge.

Split

One batch feeds multiple products, co-products, losses, and inventory states.

Identity change

Ore becomes precursor, cells become modules, grain becomes packaged food.

The representation alone is not novel: EPCIS already defines many-input/many-output TransformationEvents, and weighted Petri-net traceability has modeled split/merge processes.[4][31] The research burden is dependent contradictory evidence, joint identified query regions, tractable inference, and witness certificates on typed identity-changing histories.

Conservation without accounting fiction

Mass can be conserved through declared sinks and sources. Carbon footprints, monetary value, avoided emissions, and social attributes obey different semantics and must not be mixed.

Material balance for event e:
$S_e x_e + u_e = 0, \quad x_e \geq 0, \quad u_e$ records measured inputs, losses, emissions, or recovered outputs

Query-specific impact attribution:
$c_{out} = A_{\nu,e}\, c_{in} + \text{processImpact}_e,$ with allocation rule ν and boundary B attached

Quantity	Physical rule	Accounting rule	Forbidden shortcut
Material	typed mass/stoichiometric balance	unit conversion and measurement uncertainty	creating recycled content twice
Energy	consumption and carrier conversion	factor location/time/version	treating certificates as physical electrons
Carbon	not a conserved product attribute	attributional propagation under ν and B	subtracting avoided emissions from inventory
Waste	material disposition and recovery yield	classification and lifecycle stage	calling unknown destination “recycled”
Claims	no physical conservation	issuer, evidence, and ownership rules	signature equals physical truth

Circularity semantics: the time-unrolled event history remains acyclic. Recovered material returns to an earlier *material class* through later events; it does not create a causal edge back in time.

NO BURDEN-SHIFTING CLAIM

A carbon improvement is not automatically sustainable. Report the full declared impact vector, service, waste, recovery, and boundary sensitivity. LANTERN can expose hidden trade-offs; it cannot choose societal values by itself.

07 · EVIDENCE MODEL

Every number carries its evidence chain

A cryptographic signature establishes issuer and integrity. LANTERN additionally models whether heterogeneous records can jointly support a physical history.

Who
issuer, facility, verifier, custodian, access controller

What
event, quantity, factor, method, claim, document hash

When
event time, record time, validity interval, version

How
sensor, invoice, calculation, declaration, proxy, audit

Quality
primary-data share, precision, representativeness, verification

Access
public, bilateral, aggregate-only, selectively disclosed

Robust observation likelihood

$$p(Y \mid h, D) \propto \prod_{g \in \text{dependency clusters } D} \Psi_g(Y_g, h, \text{method}_g, R_g)$$

D records copying and derivation among invoices, declarations, sensors, and PCFs; Ψ mixes honest and contamination states without pretending copied sources are independent.

Inconsistency localization

When the feasible set is empty, solve for the smallest or lowest-cost set of evidence relaxations that restores feasibility; report alternatives rather than declaring one guilty record.

Dependency-aware recomputation

Copied declarations, invoices derived from the same ERP record, and PCFs sharing a factor are clustered. A version update marks every dependent answer without double-counting the records as independent evidence.

TRUST MODEL LIMIT

If all upstream sources collude and no independent physical or aggregate constraint exists, truth is not identifiable. The correct scientific output is an impossibility or sensitivity statement—not “AI detects fraud.”

Answers are projections of one attainable joint region

Define the evidence-consistent set **F(Y)** and a registered query vector **Q=(q₁,…,qₘ)**. LANTERN computes the attainable region of answers produced by the *same* history—not merely unrelated marginal intervals.

$$I_Q(Y) = \{ (q_1(h),\ldots,q_m(h)) : h \in F(Y) \}$$

Marginal bounds are projections of I_Q; every endpoint carries a witness history. Endpoints from incompatible histories may never be assembled into one “best/worst” passport.

Query family	Example	Certificate
Lineage	possible/must-be ancestors and contribution fractions	supporting paths and unresolved alternatives
Impact	product carbon or PEF vector under ν, B	interval, factor provenance, allocation sensitivity
Recall	minimum, posterior, and worst-case exposed set	ancestry path and coverage-risk curve
Loss	where unexplained mass or spoilage could occur	minimal feasible sink set and quantities
Circularity	recycled content and feasible second-life routes	cycle-resolved ancestry; no double claim
Audit	next document, meter, sample, or counterparty query	expected decision/query uncertainty reduction per cost

TWO UNCERTAINTY LAYERS

Identification uncertainty is what the evidence cannot distinguish even with a perfect model. **Statistical uncertainty** comes from finite data and learned parameters. Report both—and preserve their joint dependence; more compute cannot erase the first.

09 · PARTIAL IDENTIFICATION

Sharp regions are a feature, not a weakness

The scientific question is often not “predict the missing event,” but “which joint answers are attainable across physically and evidentially admissible completions?” Uncertain databases already motivate possible worlds and min/max query semantics; LANTERN must add new structure and algorithms rather than rename them.[32][33]

T1 · Point-identifiability condition

Characterize when the observation operator and typed transformation constraints eliminate every feasible perturbation that changes the query.

T2 · Sharp joint region

For a declared linear/convex transformation class, characterize the attainable joint region for lineage, mass, and allocated-impact queries —not only unrelated marginal extrema.

T3 · Certificate duality

Return a dual witness showing which observations and constraints determine each bound; unsupported precision becomes visible.

T4 · Approximate inference coverage

Bound empirical interval under-coverage by optimization error, learned-likelihood error, and constraint misspecification diagnostics.

Query identifiability at h^*: $d \in T_{F(Y)}(h^*)$ and $Dq(h^*)d \neq 0$ must imply d is ruled out.
Equivalently for a linearized class: $\ker(A_{obs}) \cap \ker(S_{typed}) \cap T_C(h^*) \subseteq \ker(Dq(h^*))$.

What must be new

- query-specific identification on **typed split/merge transformations**, not a generic graph-flow rank lemma;
- aggregate and selectively disclosed observations that mix lots and facilities;
- uncertain yields, losses, co-products, and versioned allocation rules;
- sharpness and certificates for several reconciled query families.

KILL GATE

If the theorem is merely “full column rank implies unique recovery,” if sharpness is defined tautologically without an algorithm/complexity result, or if FlowSymm and constrained open-world databases already supply the result, the theory is not a flagship contribution.[9][33]

No green answer may contradict another green answer

Product, facility, supplier, firm, waste, and circularity views must reconcile because they are computed from the same history and explicit allocation semantics.

For every feasible history h and compatible query partition P:
$q_{parent}(h; \nu, B) = \Sigma_{k \in P}\, q_{child,k}(h; \nu, B) + residual_P(h; \nu, B)$
residual is explicit: inventory change, loss, co-product allocation, boundary exclusion, or measurement error—not a hidden patch.

Joint coherence by construction

All answers share a witness history. A product-footprint endpoint cannot be combined with a recall endpoint from an incompatible world; the attainable joint region preserves their dependence.

Approximate coherence monitor

When amortized inference is used, project samples onto the feasible set or report reconciliation residuals and a bound on induced query error.

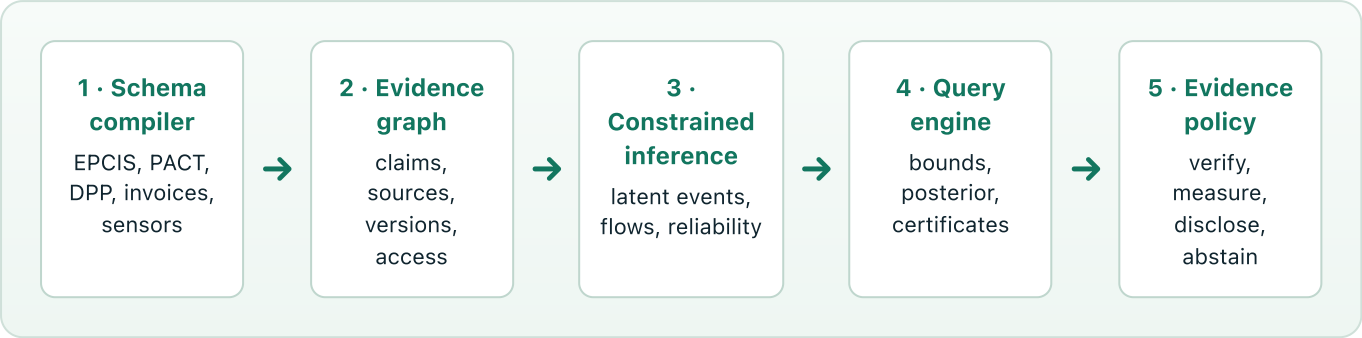
Reconciliation test	Expected invariant	Failure meaning
lot → product → facility mass	inputs = outputs + inventory + declared loss	missing event, unit error, or leakage
product PCFs → facility period total	sum plus allocation residual matches meter/factor view	boundary or allocation inconsistency
recycled inputs → recovered outputs	cycle-resolved mass not claimed twice	double counting or unknown origin
recall set → lineage posterior	coverage and false-recall risk derive from same ancestry	separate brittle rule engine

KEY PAPER CLAIM

LANTERN does not promise that all firms report the same inventory boundary. It promises that, under a declared boundary and allocation rule, all answers generated from one passport are mutually reconcilable and their disagreements are attributable.

11 · SYSTEM ARCHITECTURE

A research engine, not a dashboard stack



Scientific versus infrastructure components

Component	Paper contribution	Engineering support
Schema layer	typed semantics needed by the formal problem	connectors, OCR, LLM extraction, data cleaning
Inference	constraint-aware posterior and partial-identification engine	distributed storage and serving
Queries	coherence theorem, sharp bounds, provenance certificates	UI and reporting templates
Evidence policy	query-specific acquisition objective and guarantees if included	workflow tickets and audit scheduling
Standards views	demonstrate compatibility and version sensitivity	regulatory forms; never claim legal compliance automation

LLM BOUNDARY

An LLM may extract candidate events or compile a user question into a typed query. It cannot change mass balances, select an allocation rule silently, or certify its own extraction. All executable answers pass deterministic schema, evidence, and feasibility checks.

Amortize the posterior; never amortize away the constraints

A scalable model proposes likely latent histories, while a declarative feasibility layer enforces types, units, transformation grammar, and evidence semantics.

```
Training objective = evidence likelihood + posterior calibration + constraint projection
+ query-bound supervision
L = −E_{qψ(h|Y)}[log pφ(Y|h) + log pθ(h)] + β·KL + λ·violation(h) + γ·queryCoverageLoss
```

Neural proposal

A temporal hypergraph encoder handles variable topology, event types, mixed evidence, and learned source/noise embeddings. It proposes missing events, continuous quantities, and lineage hypotheses.

Declarative corrector

Linear, mixed-convex, or differentiable constraints project proposals onto feasible histories. Exact optimization computes decisive query bounds and dual certificates.

Posterior sampler

Sequential Monte Carlo, constrained variational inference, or diffusion over event sets represents multimodal histories; probability calibration is evaluated separately from feasibility.

Query compiler

A typed domain-specific language turns “footprint,” “recall,” or “recovery” requests into functions over the same history, allocation version, and lifecycle boundary.

Inference modes

Mode	Return	When allowed
Exact bound	provably sharp interval and dual witness	linear/convex declared subclass
Certified relaxation	outer interval with relaxation gap	nonconvex or integer transformation choices
Posterior estimate	samples, credible region, calibration report	learned stochastic components; never called identified
Conflict mode	minimal evidence/constraint conflict sets	no feasible history under current assumptions

COMPARISON DISCIPLINE

Match compute and information. Compare the neural proposal alone, projection alone, FlowSymm-like conservation models, generic hypergraph networks, and exact small-instance inference. Gains must come from transformation/evidence structure—not simply a larger model.

13 · DECISION-CRITICAL EVIDENCE

Ask for the minimum evidence that can change the answer

When an interval is too wide, LANTERN should not request “more data” in general. It should rank the document, meter, sample, or selective disclosure that most improves a declared query or decision.

$$\text{Choose } m^* = \operatorname{argmax}_{m \in M} \{ E[\, U(F(Y)) - U(F(Y \cup \text{outcome}(m))) \,] / \text{cost}(m) \}$$

U can be query-interval width, worst-case recall size, false-compliance risk, or downstream regret—not generic entropy alone.

Document
request a missing invoice, bill of material, or transformation record

Measurement
meter energy, weigh material, inspect temperature, sample composition

Verification
audit one claim or source with an independent procedure

Disclosure
request a safe aggregate, range, proof, or redacted subgraph

Link query
confirm whether a specific lot/facility relation exists

Stop
certify that no affordable query can alter the intended decision

Novelty boundary

Active feature acquisition and cost-sensitive sensing are established fields.[10] LANTERN earns this extension only if acquisition acts on **structured transformation evidence**, targets **sharp identified sets or coherent decisions**, and respects **selective disclosure and verification costs**.

RECOMMENDED SCOPE

Keep active evidence acquisition as a secondary but compelling result. The flagship claim remains identification and coherence. If the acquisition theorem becomes generic, present it as an experiment—not a second paper bolted onto the first.

Visibility without demanding the whole supplier graph

The goal is not to reconstruct commercially sensitive relationships at any cost. The research question is which sustainability queries remain identifiable under permitted disclosures.

Disclosure operators

- facility or period aggregates;
- intervals and moments;
- redacted subgraphs;
- secure sum or zero-knowledge claim;
- federated likelihood/gradient updates;
- selective verifiable credentials.

Threat model

- relationship and volume inference;
- membership inference;
- honest-but-curious coordinator;
- strategic under-disclosure;
- colluding suppliers;
- differencing attacks across repeated queries.

Privacy–identifiability frontier

Characterize the coarsest disclosure operator D for which a specified query q is point- or set-identified under typed transformations.

Leakage–width trade-off

Bound how query interval width changes as aggregation, noise, or redaction removes identifiable directions.

```
q-identifiable under disclosure D iff every h₁,h₂ ∈ F with D(h₁)=D(h₂) also satisfy
q(h₁)=q(h₂).
Otherwise return the sharp disclosure-induced interval—not a privacy-blind point
estimate.
```

NOT SOLVED BY FEDERATION ALONE

Federated learning keeps raw data local but can still leak information and does not make heterogeneous graphs, missing events, or false claims identifiable. Privacy evaluation must include attacks, communication, utility, and repeated-query composition.

One history, multiple disciplined views

LANTERN separates quantities that are often collapsed: physical flow, attributional inventory, consequential change, circular claim, and communicated ownership.

View	Question	Primary output	Guardrail
Physical	what material/energy moved and transformed?	typed flow history and losses	mass and unit feasibility
Attributional	what footprint is assigned under ν,B?	PCF/impact vector plus uncertainty	method/version attached
Consequential	what changes if action a occurs?	counterfactual delta, if separately identified	never inferred from inventory alone
Circular	what came from and returns to another cycle?	cycle-resolved origin and recovery	no double counting or free avoided emissions
Claim	who may communicate or own a reduction?	evidence and contractual status	legal/normative layer remains external

Standards-aware outputs

PACT-compatible PCF view

Populate and validate declared fields while attaching evidence and inference status; compatibility is not verification.

DPP view

Expose role-appropriate data, identifiers, provenance, and query certificates through the relevant product schema.

PEF / multi-impact view

Report the declared lifecycle impact vector to reveal burden shifting beyond a single carbon score.

STRONG BUT HONEST UMBRELLA

The paper can demonstrate carbon, waste, circularity, and recall because these are transformations or queries on the same history. It should not claim to solve causal intervention effects unless a separate identification design supplies them.

16 · BENCHMARK THESIS

SC-PASSPORT-Bench

A controlled benchmark where the complete physical history is known, only fragmented enterprise evidence is released, and every corruption has an explicit causal origin.

Ground truth
complete event, material, impact, loss, and return histories

Observed layer
EPCIS-like events, invoices, sensors, PCFs, aggregates, credentials

Corruptions
missing, duplicate, conflicting, strategic, stale, aggregate-only

Queries
lineage, footprint, recall, loss, recovery, conflict, next evidence

Four benchmark tracks

Track	Purpose	What is hidden	Headline evidence
A · Exact synthetic	theorem and algorithm verification	controlled transformations/evidence	sharpness against oracle enumeration
B · Calibrated semi-synthetic	realistic topology, yields, factors, missingness	injected events and errors	robustness across sectors
C · Real replay	external validity on partner/open event traces	pre-registered holdouts	recovery and human-audited conflicts
D · Privacy challenge	selective disclosure frontier	tiers, edges, quantities, identities	query utility versus leakage

ANTI-TOY REQUIREMENT

Synthetic ground truth is necessary but insufficient. At least one real or semi-real trace must preserve actual event timing/topology and be evaluated by domain reviewers. Results must survive strategic missingness, not only random masking.

Cold food: perishability makes missing history consequential

Transformation grammar

harvest → grading → blending/processing → packaging → pallet aggregation → cold storage → distribution → sale/expiry → donation, feed, compost, digestion, or disposal

Evidence

lot IDs, EPCIS events, temperature intervals, inventory snapshots, yield, shelf-life, transport mode, electricity/fuel factors, waste disposition

Research tasks

Task	Ground truth	Decision relevance
hidden blending and split recovery	lot-level parent fractions	targeted recall rather than blanket disposal
cold-chain anomaly localization	temperature exposure and spoilage mechanism	separate physical risk from missing sensor data
food-loss reconciliation	mass and disposition by stage	locate preventable loss without inventing yield
product carbon interval	energy, transport, waste, allocation	quantify uncertainty from missing primary data

Public calibration assets

Use GS1 fresh-food traceability examples and EPCIS semantics, FAO food-loss data, UNEP Food Waste Index, public emission factors, and FreshRetailNet-50K only for demand/perishability context—not as a fake traceability ground truth.[11]

INDIA–SWITZERLAND DEMONSTRATOR

A temperature-controlled ingredient or specialty food network can connect production, international transport, storage, and recovery. Country labels are context; the method must generalize across topology and product.

Textiles: blending and subcontracting stress lineage

Transformation grammar

fiber source → spinning/blending → weaving/knitting → wet processing → cutting → assembly → distribution → rental/resale → sorting → fiber recycling

Evidence

fiber declarations, mass tickets, purchase orders, subcontractor events, chemical/energy records, product labels, RFID scans, sorting and recycling yields

Hard cases

Blend ambiguity
several fibers and lots contribute to one fabric roll

Hidden subcontractor
a process exists without disclosed supplier identity

Mass loss
cutting scraps and wet-process changes are inconsistently recorded

Recycled claim
origin, cycle, and allocation can be duplicated

Rental loop
condition and cleaning events alter routing and impacts

Selective disclosure
supplier relationships are commercially sensitive

Open Supply Hub can seed facility identities and broad relationships, while DPP case studies and public RFID/rental traces can calibrate process timing. Neither supplies complete material ground truth; the benchmark must label what is observed, licensed, simulated, and partner-provided.[12]

EXTERNAL-VALIDITY TEST

Ask independent textile experts to review transformation grammars, missingness patterns, allocation choices, and injected fraud. A benchmark designed only by ML researchers risks encoding physically easy but operationally irrelevant errors.

Batteries: the first mandatory DPP stress test

The EU battery passport becomes mandatory for relevant categories from 18 February 2027, making batteries a timely but collision-sensitive sector.[13]

Transformation grammar

raw material → refining → active material → cell → module → pack → vehicle/storage use → diagnosis → second life → dismantling → black mass → recovered material

Evidence

material origin, chemistry, batch genealogy, energy mix, cell test, pack assembly, state of health, transport, dismantling, recovery mass and quality

Evaluation questions

Question	Why existing passport data are insufficient	LANTERN output
Which packs contain suspect precursor?	unrecorded blending and split events	probabilistic and worst-case recall sets
Is recycled content physically feasible?	aggregate claims can be duplicated across packs	cycle-resolved mass interval and conflict witness
What is cradle-to-gate carbon?	facility energy and product lineage are both uncertain	method-versioned PCF interval and sensitivity
Reuse or recycle?	identity and condition evidence may conflict	feasible routing set with evidence gaps

Argonne R&D GREET can calibrate battery lifecycle inventories; the European Commission battery-passport data points define relevant fields. They do not provide lot-level histories, so event ground truth must be generated or obtained from a partner and labeled accordingly.[14]

Real failures are structured, correlated, and strategic

Family	Generator	Severity axis	What it tests
Random missing	independent event/field mask	5–80%	basic completion; easiest track
Tiered missing	probability rises with depth/SME status	tier and digital maturity	realistic visibility decay
Strategic omission	hide high-impact, low-yield, or noncompliant events	selection strength	MNAR robustness and false confidence
Aggregate disclosure	replace lot events with facility/month totals	aggregation grain	privacy–identifiability frontier
Conflict	invoice, sensor, meter, and declaration disagree	noise and source correlation	conflict localization
Identity error	duplicate IDs, wrong lot, hidden split/merge	frequency and path depth	lineage multimodality
Method drift	factor, boundary, unit, or allocation version changes	version distance	semantic robustness
Adversarial claim	feasible-looking recycled or low-carbon fabrication	attacker knowledge	limits of physical evidence
Churn	supplier/product/topology enters or exits	OOD graph shift	generalization and abstention

Pre-registered corruption matrix

Train on a subset of corruption types and severities; test on held-out combinations. A headline result must include **strategic missingness + aggregate disclosure + method drift**, not merely random event deletion.

> **NO BENCHMARK LEAKAGE**
>
> Hidden histories, corruption seeds, and test partner mappings remain sequestered. LLM extraction tools may see only observed evidence, never generator templates or hidden event descriptions.

Score truth, uncertainty, coherence, and usefulness

Task	Primary metrics	Critical failure metric
Lineage recovery	edge/path precision-recall, contribution error, ancestry calibration	false certainty under strategic missingness
Flow completion	quantity MAE, typed feasibility, mass residual	physically impossible sample rate
Footprint bounds	coverage, mean/median width, sharpness gap, factor sensitivity	under-coverage and hidden boundary drift
Recall	coverage–waste curve, exposed-set precision, worst-case size	missed affected product
Conflict localization	AUPRC, minimal-set recovery, human resolution time	blaming an honest source under collusion
Circular claims	recycled-content interval, double-count detection, recovery feasibility	cycle duplication accepted as valid
Evidence acquisition	interval/regret reduction per cost, verified coverage, stopping quality	excess burden on small suppliers
Privacy	utility versus ε/leakage, membership/link attack success, communication	sensitive relation reconstruction
Coherence	cross-query reconciliation residual and violation rate	each KPI looks accurate but totals disagree

PRIMARY ENDPOINT

Coverage–width–coherence frontier under structured missingness. Point accuracy alone rewards unjustified certainty. The best method returns the narrowest valid intervals while keeping every answer physically and accounting coherent.

Beat the strongest explanation, not a weak GNN

Baseline family	Representative capability	LANTERN must demonstrate
Rules + joins	deterministic EPCIS traversal and reconciliation rules	calibrated alternatives when links/events are missing
LP / matrix balancing	mass-balance completion, min-cost feasible flow	typed identity, multimodality, evidence reliability, query bounds
FlowSymm / physics GNN[9]	learned missing-flow completion with exact conservation	split/merge transformation advantage and sharp non-identifiability
Temporal graph/hypergraph models	link/event/quantity imputation	feasibility, provenance, coherent cross-query uncertainty
Probabilistic databases[17]	provenance and probabilistic query evaluation	learned physical transformation constraints and identified sets
DPP / digital twin systems[15]	data containers, interoperability, lifecycle applications	uncertain reconstruction and theory—not another architecture diagram
Robust Bayesian models	source-specific noise and contamination	structured conflicts linked to transformation feasibility
Independent task models	separate best lineage, PCF, recall, waste predictors	comparable accuracy with near-zero reconciliation error
Oracle programs	complete-history truth and exact bounds	small-instance correctness and quantified approximation gap

Capacity and information matching

Match hidden dimension, parameter count, evidence fields, training compute, and solver budget. Give deterministic baselines the same transformation grammar. Otherwise the result only shows that semantics help a baseline denied those semantics.

PRIMARY COMPARISON

FlowSymm-style conservation completion + a post-hoc query engine is the most dangerous baseline. LANTERN must win because typed transformations and identified-set reasoning matter, not because it has a better encoder.

Ten falsifiable studies

ID	Study	Primary endpoint	Failure interpretation
E1	small-instance oracle enumeration	bound sharpness and posterior calibration	engine is not computing what theory claims
E2	identifiability phase diagram	coverage/width versus observation rank, type, and placement	theoretical boundary does not predict practice
E3	structured missingness sweep	coverage–width–coherence frontier	works only under random masking
E4	hidden split/merge and identity change	query error versus FlowSymm and generic hypergraphs	transformation structure adds no value
E5	conflict and strategic-claim localization	minimal-set AUPRC and false accusation rate	robust likelihood is cosmetic
E6	cross-query stress	product/facility/firm/waste reconciliation residual	one-posterior claim fails operationally
E7	active evidence budget	decision/query uncertainty reduction per cost	generic heuristics acquire equally useful data
E8	privacy/disclosure frontier	interval width, attack success, communication	privacy makes answers useless or leaks relations
E9	sector/topology/version OOD	calibration, abstention, reconciliation	learned model memorizes one schema
E10	real or enclave replay	expert-validated conflicts, query coverage, runtime	synthetic success does not transfer

Required ablations

no type system · **no transformation constraints** · **no source model** · **no provenance** · **point-only** · **independent query heads** · **no projection** · **generic entropy acquisition** · **full vs aggregate disclosure** · **fixed vs drifting method version**

Make false confidence the main error

Pre-registration

Freeze sector grammars, corruption families, hidden-query sets, primary endpoints, baseline information, and partner evaluation before training.

Paired evaluation

Use identical latent histories and corruption seeds across methods; bootstrap at the history/network level rather than treating every event as independent.

Calibration

Reliability diagrams, interval coverage by corruption/tier/query, proper scoring rules, selective-risk curves, and expected calibration error with uncertainty.

Sharpness

Report width only conditional on valid coverage. Compare exact identified intervals on small cases and relaxation gaps on large cases.

Multiple seeds

Five or more training seeds, generator seeds separated from model seeds, paired confidence intervals, and all failed/infeasible runs retained.

Human validation

Blind experts assess conflict sets and evidence requests; report agreement, time, false accusation, and whether the suggestion changed the resolution.

Primary result panel

Validity
nominal 90/95% query coverage

Precision
conditional interval width

Consistency
reconciliation residual

Utility
recall waste / audit cost / decision regret

REPORTING PROHIBITION

Do not average away non-identifiable cases. Report point-identified, partially identified, conflicted, and abstained subsets separately, including their sector and tier distribution.

The results that could make this ICLR-level

Theorem A · Query identifiability

Necessary and sufficient conditions for a query to be invariant over typed split/merge histories under lot-level and aggregate observations. Must expose the extra role of transformation types and lineage identity.

Theorem B · Joint region and witnesses

For a nontrivial subclass with uncertain yield/allocation, characterize the attainable multi-query region and return histories/dual evidence witnessing exposed boundaries.

Theorem C · Coherence under approximation

Exact samples are coherent by construction; for projected amortized inference, bound every declared reconciliation error and the induced query distortion by projection and solver residuals.

Theorem D · Disclosure frontier

Characterize the minimum aggregate/statistical disclosure needed to identify q, and prove a lower bound showing when any method must return a wide interval.

Theorem E · Complexity and tractability

Classify query/region complexity by transformation arity, treewidth, aggregation, cycles/rework, and allocation; give exact or approximation algorithms for a useful tractable fragment.

Impossibility · Colluding evidence

Without independent constraints or disclosure, construct indistinguishable physical histories with different sustainability answers. This justifies abstention and prevents fraud-detection hype.

Scope discipline

Result	Main paper?	Reason
A + B	**Yes · non-negotiable**	scientific nucleus
C	**Yes**	connects theorem to scalable system
D	Yes if clean	important multi-enterprise novelty
E	**Yes**	prevents “possible-world min/max” from being a renamed database result
Impossibility	Yes	honest boundary and strong motivation

What is already taken—and the surviving wedge

Prior family	Already established	LANTERN must add
Conservation flow completion[9]	missing network flows with physical constraints and symmetry	identity-changing hyperedges, aggregates, sharp query regions, evidence provenance
EPCIS / weighted Petri nets[4][31]	typed many-input/many-output events, split/merge, quantities, sensor traceability	representation is infrastructure; novelty must be uncertainty semantics and inference
Uncertain/inconsistent databases[32][33]	possible worlds, lineage, repairs, min/max probabilistic queries under constraints	joint quantitative region, witnesses, complexity/tractable fragments for physical transformations
Bayesian material-flow analysis[34]	uncertain, missing, contradictory, and over-aggregated material reconciliation	lot/item lineage plus joint carbon/waste/recall queries and non-Bayesian identified limits
Probabilistic provenance[17]	lineage/probability of database query answers	learned physical histories and transformation feasibility
Privacy-preserving carbon claims[18]	cryptographic or selective proof of emissions claims	truth under incomplete/conflicting physical evidence and query identification
DPP architectures/reviews[15][16]	containers, interoperability, adoption, lifecycle use cases	formal uncertain reconstruction and coherent executable query semantics
Predictive/dynamic passports[19]	LLM assistants, condition updates, routing or circular decisions	identified-set theory and multi-query consistency
Federated visibility models[20]	link prediction across data silos	heterogeneous transformations, selective disclosure frontier, downstream query validity
Provenance privacy[35]	formal privacy–utility trade-offs from hiding workflow provenance	query identifiability/leakage for material, carbon, waste, and repeated disclosure
Active feature acquisition[10]	cost-sensitive observation selection for prediction/decision	structured evidence requests targeting physical identified sets

Two especially direct 2026 systems collisions already provide carbon-enriched BOM graphs and governance-aware event traceability with selective disclosure.[37][38] Architecture integration is therefore not the novelty.

DEFENSIBLE HYPOTHESIS—NOT A "FIRST" CLAIM

LANTERN defines and computes a joint attainable region—with witness histories—for lineage, impact, loss, recall, and circularity queries on uncertain typed transformations, while preserving dependent evidence and disclosure semantics.

An exact-name web collision check through 30 August 2026 found no relevant research use of "LANTERN-SC"; this is a naming check, not proof of scientific novelty.

Why this is not “digital twin + agents + carbon”

That direction now has a direct 2026 collision: cooperative digital twin–MARL already combines planning, inventory, logistics, expediting, recycling, cost, CO_2, waste, service, and shocks.[21]

Generic super-platform	LANTERN-SC	Reviewer interpretation
many agents/modules	one posterior over physical history	integration versus scientific object
weighted cost/carbon/waste reward	query semantics and explicit boundaries	preference engineering versus identification
simulator state assumed known	state itself is incomplete and conflicting	control benchmark versus evidence problem
dashboard explanations	dual/provenance/reconciliation certificates	presentation versus falsifiable evidence
actions from learned policy	answers/abstention from identified set	autonomy versus epistemic discipline

ISOMORPH also supplies a configurable public supply-chain digital twin, so “first open supply-chain twin” is unavailable.[22] LANTERN may use a twin as a generator, but the twin is not the contribution.

DO NOT DILUTE

Adding procurement agents, routing optimization, negotiation, forecasting, and an LLM copilot would make the proposal look broader and the paper weaker. Those are consumers of an executable passport, not core novelties.

Three proposals, three different questions

Proposal	Core question	Learned object	Primary evaluation
MOSAIC-SC	Which mechanisms changed under a disruption, and can known atoms compose?	local transition mechanisms and intervention masks	zero-transition compound-shock recovery
RSSL	Can one compact scenario law serve a declared family of future optimization programs?	specification-independent conditional scenario law	worst-program regret, feasibility, and support size
LANTERN-SC	What physically happened, which evidence supports it, and what remains unknowable?	posterior/identified set over transformation histories	coverage, sharpness, coherence, and evidence utility

Potential interfaces—without merging papers



PORTFOLIO LOGIC

LANTERN is the most direct “one-stop sustainability evidence” idea. MOSAIC has the highest disruption-generalization ambition; RSSL has the cleanest stochastic-programming theory. They may share data infrastructure while preserving separate novelty claims.

Questions the paper must answer before submission

“Just flow completion?” — fatal risk

Show a theorem and experiments where typed split/merge identity, aggregate evidence, and query sharpness change the answer beyond ordinary conservation.

“All synthetic?” — fatal risk

Provide real/enclave replay, domain-reviewed corruption, and calibration from trace/event assets.

“Intervals useless?” — high

Map width to recall waste, audit cost, or decision regret; demonstrate evidence that meaningfully closes it.

“Standards project?” — high

Keep connectors in the artifact; the method/theory must stand without regulatory novelty.

“Prior provenance work?” — high

Compare with probabilistic databases and articulate physical-history learning versus tuple provenance.

“Privacy hand-wave?” — high

Specify adversary, disclosure operator, leakage metric, and impossibility boundary.

“Too many contributions?” — medium

Make identifiability + sharp coherence the spine; evidence acquisition and standards are consequences.

CLAIMS TO REJECT

- not the first DPP
- not fraud-proof
- not universal ESG
- not causal impact by default
- not a legal compliance oracle
- not a blockchain paper

Stop conditions protect the research

Kill test	Threshold / evidence	Response
Theory collapse	main condition is ordinary rank/nullspace with renamed variables	reformulate around query-specific transformations or stop
Vacuous uncertainty	realistic evidence leaves most headline intervals decision-useless	narrow sector/query or make evidence acquisition primary
No structural gain	FlowSymm/generic hypergraph matches all core endpoints	reject flagship claim
Random-only success	coverage/width collapses under strategic or tiered missingness	fix missingness model or stop
Synthetic-only	no partner/enclave/replay validation and no credible domain audit	target benchmark/workshop, not flagship method paper
Coherence cost	constraints destroy predictive performance or scalability	publish boundary/relaxation if scientifically new; otherwise stop
Privacy failure	useful queries reveal sensitive links or private setting is useless	exclude or turn into explicit impossibility result
Platform drift	work becomes connectors, LLM extraction, dashboards, or compliance forms	separate engineering demo from research paper

Minimum flagship bar

A nontrivial theorem, valid and meaningfully narrow intervals under structured missingness, decisive advantage on hidden transformations, near-zero cross-query inconsistency, and one credible external validation.

What can be built openly—and what cannot

Asset	Use	Caveat
GS1 EPCIS 2.0 examples[4]	event schema and transformation semantics	standard/examples, not ground-truth industrial history
PACT v3 / iLEAP[2]	PCF and logistics fields, quality/version semantics	audit license and evolving technical versions
Open Supply Hub[12]	facility skeletons, geocoding, broad relations	API/license limits; no lot-level flow truth
UN Comtrade / PortWatch[23]	aggregate trade/topology/shock calibration	country/commodity aggregation
EPA USEEIO / UK factors[24]	secondary impact priors and sensitivity	sector averages are not supplier primary data
FAO / UNEP food loss[11]	loss-rate and disposition calibration	heterogeneous methods and geography
Argonne GREET[14]	battery lifecycle inventory calibration	model terms/version must be frozen
Partner enclave	real transformations, evidence gaps, expert adjudication	no raw release; publish schema, protocol, aggregates, and code

Data cards must separate four statuses

observed open · observed restricted · synthetic calibrated · hidden ground truth

LICENSE RULE

“Publicly accessible” does not mean redistributable. Record source, version, license, access date, API limits, permitted transformations, and whether derived event histories may be released.

An artifact reviewers can interrogate

Schema registry

Versioned lot, material, event, evidence, allocation, boundary, and query types with validators and migration tests.

History generator

Sector grammars, topology templates, physical parameters, hidden histories, corruption seeds, and deterministic manifests.

Exact evaluator

Small-instance enumeration/optimization, sharp bound certificates, conflict oracles, and cross-query reconciliation tests.

Model package

Encoder, proposal distribution, constraint projector, posterior sampler, query compiler, and evidence policy.

Evidence cards

For every derived answer: sources, versions, quality, verification, assumptions, feasible alternatives, and dependency graph.

Evaluation cube

Per-history × corruption × query × method × seed outputs, including abstentions, failures, runtime, and memory.

Reviewer tests

- change one emission factor and identify every affected answer;
- delete a hidden transformation and reproduce the predicted interval expansion;
- verify that product/facility/waste views reconcile from raw events;
- run a disclosure attack and reproduce the privacy–utility curve;
- regenerate every benchmark split without seeing hidden test histories.

ARTIFACT PHILOSOPHY

The PDF can promise an open generator and evaluation protocol. It must not promise redistribution of partner data. Enclave scripts, synthetic twins, and aggregate result cards can still make the real validation auditable.

33 · PAPER BLUEPRINT

A coherent top-tier submission

Section	Scientific job	Non-negotiable evidence
1 · Introduction	passports store claims but do not infer coherent histories	one contradiction example and standards gap
2 · Problem	typed transformation hypergraph, evidence, query, identified set	explicit information/assumption contract
3 · Theory	query identifiability, sharpness, dual evidence certificates	nontrivial transformation subclass and impossibility
4 · Method	amortized constrained posterior and query engine	feasibility/scalability and approximation diagnostics
5 · Coherence/privacy	reconciliation and selective-disclosure frontier	formal property plus attack evaluation
6 · Benchmark	three sectors, corruption suite, public generator	hidden histories and pre-registered splits
7 · Results	coverage–width–coherence, conflicts, evidence value	FlowSymm/provenance/DPP baselines and ablations
8 · External validity	real/enclave replay and expert review	failures and human resolution outcomes
9 · Limits	collusion, wide intervals, legal claims, social attributes	kill tests and negative cases

One-sentence paper claim

> **We introduce a probabilistic transformation-hypergraph model that returns sharp, evidence-certified, mutually coherent sustainability queries from incomplete multi-enterprise histories, and characterize when those queries are identifiable under aggregate or selective disclosure.**

Do not put “one-stop solution” in the paper title. Demonstrate umbrella value through the query algebra and benchmark; lead with the precise scientific claim.

Conditional GO · strongest evidence-centric umbrella

LANTERN-SC has real research "damm" because it attacks the hidden assumption beneath most sustainable supply-chain AI: that the state, lineage, footprint, and evidence quality are already known.

Novelty nucleus

Typed identity-changing transformation hypergraphs + sharp query-specific partial identification + cross-query coherence + evidence certificates under aggregate and conflicting disclosures.

Decisive experiment

Under strategic missing transformations, LANTERN maintains nominal footprint/recall coverage with materially narrower intervals and near-zero reconciliation error, while FlowSymm, generic hypergraphs, independent task models, and post-hoc DPP pipelines become overconfident or inconsistent.

Fatal outcome

The main theorem is ordinary flow rank, realistic intervals are vacuous, or the advantage disappears when baselines receive the same physical grammar.

90-second pitch

Most supply-chain AI starts after the hardest epistemic problem has been assumed away. Firms often do not know the exact ancestry, material losses, footprint, or evidential reliability of the products they optimize. Digital product passports and carbon-exchange standards can carry structured claims, but a signed and syntactically valid claim can still be missing, stale, contradictory, or physically impossible. LANTERN-SC represents lots and identity-changing split/merge processes as a typed stochastic transformation hypergraph, links every observation to its source and method, and infers the set and posterior of physical histories consistent with the evidence. Lineage, product impact, recall exposure, waste localization, and circular recovery are then coherent queries on that same object. When the evidence cannot identify an answer, LANTERN returns the sharpest possible interval and a certificate explaining what controls it; when the interval is decision-useless, it requests the cheapest decisive evidence or abstains. The contribution is not a dashboard or a super-agent. It is a theory and benchmark for knowing what a sustainable supply chain can justifiably claim.

Primary sources and collision anchors

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3. EU Ecodesign for Sustainable Products Regulation 2024/1781; European Commission Digital Product Passport.
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5. UN Transparency Protocol Digital Product Passport; W3C Verifiable Credentials 2.0.
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8. GHG Protocol Scope 3 Phase 1 Progress Update, March 2026.
9. FlowSymm: Flow-Based Symmetry for Learning on Network Flows, ICLR 2026; Flow-Attentional Graph Neural Networks; Combining Physics and Machine Learning for Network Flow Estimation, ICLR 2021.
10. Valancius et al. Active Feature Acquisition with Generative Surrogate Models, ICML 2024; Li et al. Towards Cost Sensitive Decision Making, AISTATS 2025; Piskorz et al. Active Feature Acquisition for Treatment Assignment, AISTATS 2025.
11. FAO Food Loss and Waste Database; UNEP Food Waste Index 2024; FreshRetailNet-50K.
12. Open Supply Hub API documentation; Wang et al. DPP in garment rental operations.
13. European Commission, Digital Product Passport for Batteries; EU Batteries Regulation 2023/1542.
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28. OECD, Strengthening supply chains through efficiency, resilience, AI and environmental performance (2026).
29. European Commission JRC, DPP data requirements and system perspective.
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32. Benjelloun et al. ULDBs: Databases with Uncertainty and Lineage; Arenas et al. Consistent Query Answers in Inconsistent Databases; Rekatsinas et al. HoloClean.
33. Ceylan et al. Open-World Probabilistic Databases; Friedman and Van den Broeck. Constrained Open-World Probabilistic Databases.
34. Lupton and Allwood. Incremental Material Flow Analysis with Bayesian Inference (2018).
35. Davidson et al. Provenance Views for Module Privacy; Deutch et al. Optimizing the Trade-off between Privacy and Utility in Data Provenance.
36. Xie et al. Processing Uncertain RFID Data in Traceability Supply Chains (2014).
37. Carbon-enriched bill-of-materials graph framework for product carbon footprints (2026).
38. Governance-Aware, Privacy-Preserving, Event-Driven Supply Chain Traceability (2026).

FINAL LITERATURE GATE

Repeat the exact-method and exact-claim search immediately before submission. Recent 2026 papers, standards, and DPP implementation documents are moving targets; freeze access dates and versions in the artifact.